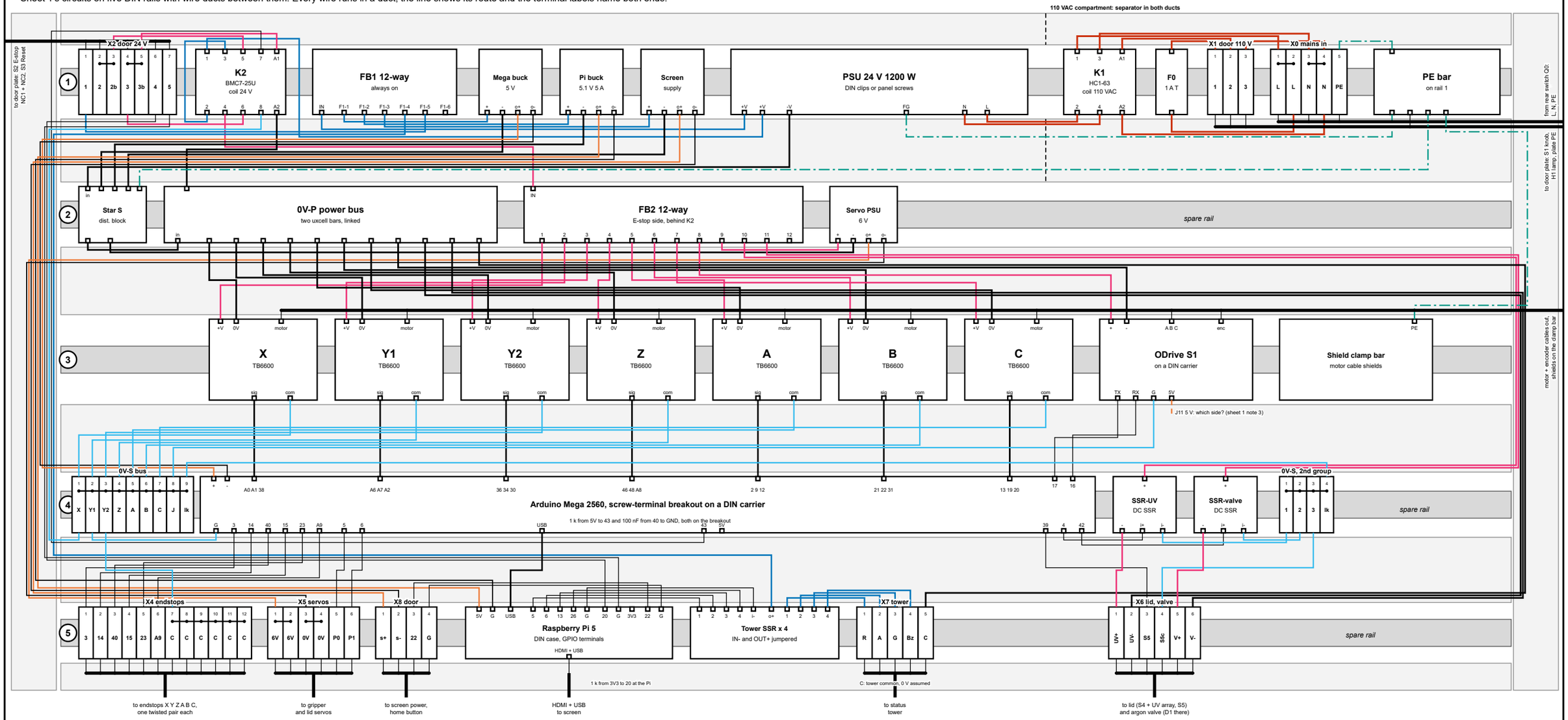
RMR control panel: DIN-rail layout and wiring

Sheet 1's circuits on five DIN rails with wire ducts between them. Every wire runs in a duct; the line shows its route and the terminal labels name both ends.



Legend

- 110 VAC
- 0 V power
- logic supply +
- multi-core signal cable
- wire duct
- +24 V always on
- 0V-S signal bus
- logic supply 0 V
- cable leaving the panel
- DIN rail (TS35)
- +24 V behind K2
- PE
- signal
- terminal block, bridged

Field cables: one twisted pair per endstop; motor cables shielded, shield to PE at the drive end.

Five rails, top to bottom

1 X2, K2, FB1, Mega / Pi / screen bucks, PSU, K1, F0, X1, X0, PE bar
2 Star S, 0V-P bus, FB2, servo PSU (spare length at the right)
3 Drivers X Y1 Y2 Z A B C, ODrive S1, shield clamp bar
4 0V-S bus, Mega breakout, SSR-UV, SSR-valve
5 Field terminals X4 X5 X8 X7 X6, Pi 5, tower SSRs

Arrangement, not to scale; terminal positions are drawn where the routing is clearest.
Land each wire on the terminal it is drawn to. Measure parts before cutting rail and duct.
Gauges, copper only: mains L and N, PSU to FB1, K2, FB2 and S, S to 0V-P: 12 AWG.
F2-1 to F2-7: 18 AWG. F2-8: 16. F0 rung: 300 V insulation. Everything else: 20 AWG.
TB6600, ODrive, bucks, SSR modules, Mega and Pi need DIN clips or carriers.
Bond every rail and the frame to PE (chassis stud); 0 V meets PE only at S.
Open items and the firmware gap: sheet 1, notes 3 to 5.

RMR control panel: DIN-rail layout

Sheet 2 of 2. Legal 14 x 8.5 in, landscape. Print at 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-14.
Status: design, not yet built or tested.
Circuits and part data: sheet 1.